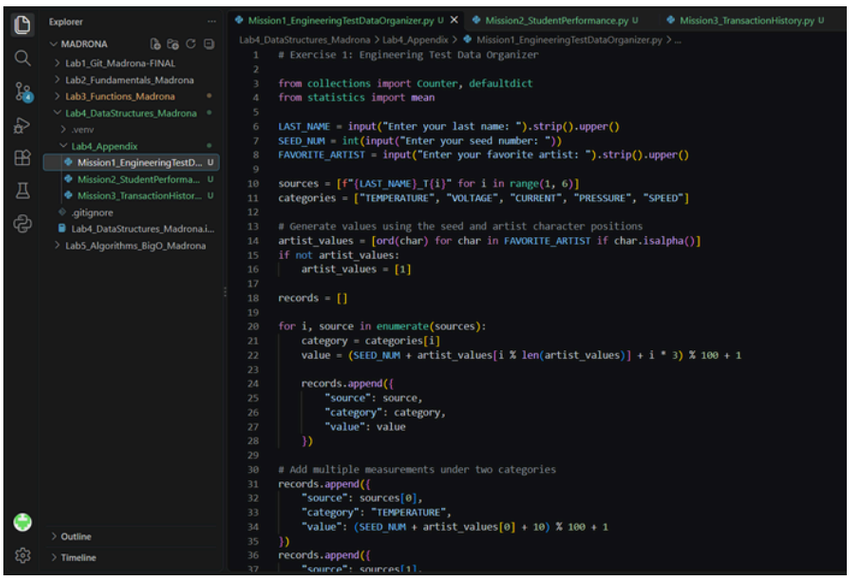
MADRONA, Glydel B.
BSECE-1C

ACTIVITY 4 - DATA STRUCTURES

Appendix 1: Engineering Test Data Organizer



```python
# Exercise 1: Engineering Test Data Organizer

from collections import Counter, defaultdict
from statistics import mean

LAST_NAME = input("Enter your last name: ").strip().upper()
SEED_NUM = int(input("Enter your seed number: "))
FAVORITE_ARTIST = input("Enter your favorite artist: ").strip().upper()

sources = [f"{LAST_NAME}_T{i}" for i in range(1, 6)]
categories = ["TEMPERATURE", "VOLTAGE", "CURRENT", "PRESSURE", "SPEED"]

# Generate values using the seed and artist character positions
artist_values = [ord(char) for char in FAVORITE_ARTIST if char.isalpha()]
if not artist_values:
    artist_values = [1]

records = []

for i, source in enumerate(sources):
    category = categories[i]
    value = (SEED_NUM + artist_values[i % len(artist_values)] + i * 3) % 100 + 1

    records.append({
        "source": source,
        "category": category,
        "value": value
    })

# Add multiple measurements under two categories
records.append({
    "source": sources[0],
    "category": "TEMPERATURE",
    "value": (SEED_NUM + artist_values[0] + 10) % 100 + 1
})
records.append({
    "source": sources[1],
    "category": "VOLTAGE",
    "value": (SEED_NUM + artist_values[0] + 20) % 100 + 1
})

# Include one repeated source/category record
records.append(records[0].copy())

# Organize records by source and category
organized = defaultdict(lambda: defaultdict(list))

for record in records:
    organized[record["source"]][record["category"]].append(
        record["value"]
    )

# Identify repeated and distinct records
record_counts = Counter(
    (r["source"], r["category"], r["value"]) for r in records
)
repeated = [item for item, count in record_counts.items() if count > 1]
distinct = set(
    (r["source"], r["category"], r["value"]) for r in records
)

# Numerical summaries by category
category_values = defaultdict(list)

for record in records:
    category_values[record["category"]].append(record["value"])

print("\n===== GENERATED TEST DATA =====")
for record in records:
    print(record)

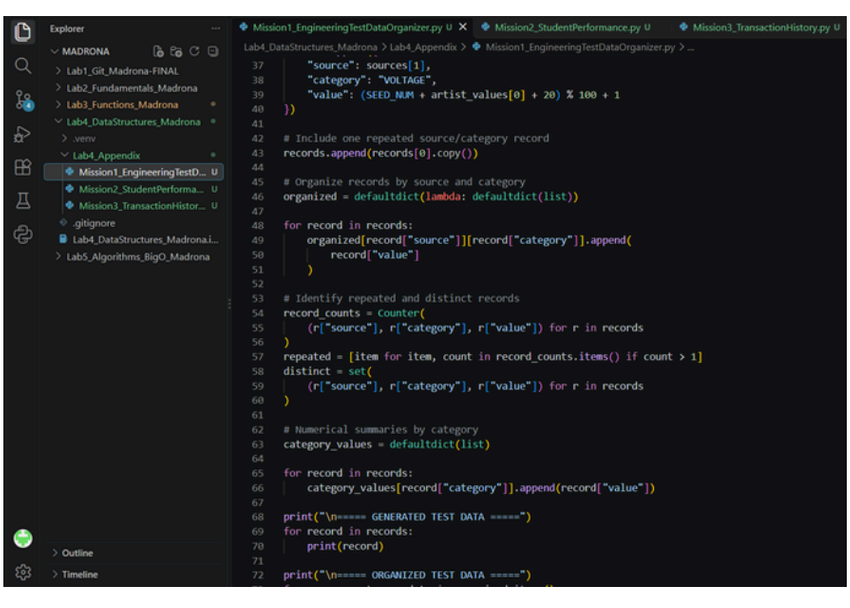
print("\n===== ORGANIZED TEST DATA =====")
for source, category_data in organized.items():
    print(source, dict(category_data))

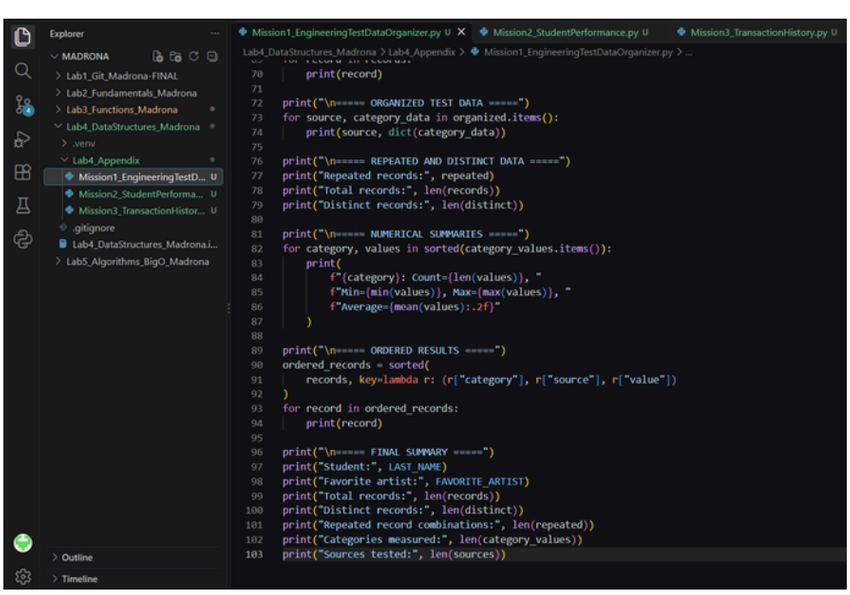
print("\n===== REPEATED AND DISTINCT DATA =====")
print("Repeated records:", repeated)
print("Total records:", len(records))
print("Distinct records:", len(distinct))

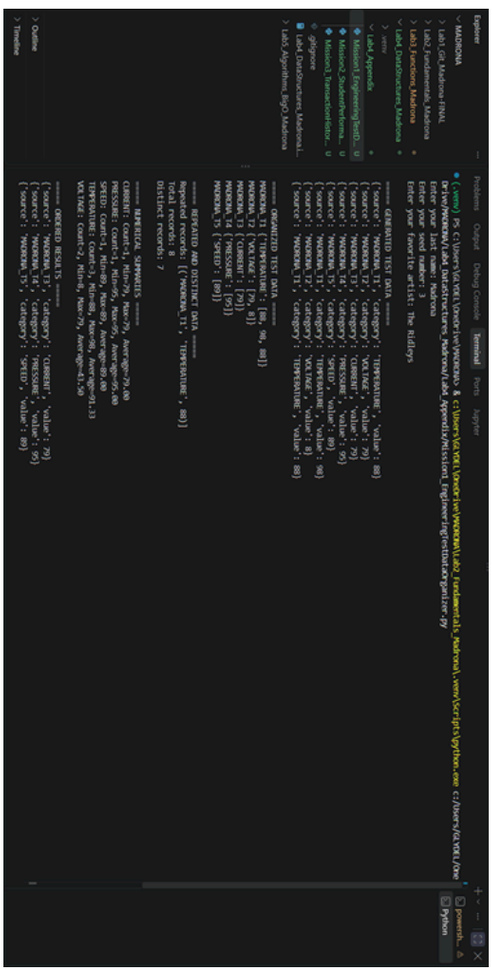
print("\n===== NUMERICAL SUMMARIES =====")
for category, values in sorted(category_values.items()):
    print(
        f"{category}: Count={len(values)}, "
        f"Min={min(values)}, Max={max(values)}, "
        f"Average={mean(values):.2f}"
    )

print("\n===== ORDERED RESULTS =====")
ordered_records = sorted(
    records, key=lambda r: (r["category"], r["source"], r["value"])
)
for record in ordered_records:
    print(record)

print("\n===== FINAL SUMMARY =====")
print("Student:", LAST_NAME)
print("Favorite artist:", FAVORITE_ARTIST)
print("Total records:", len(records))
print("Distinct records:", len(distinct))
print("Repeated record combinations:", len(repeated))
print("Categories measured:", len(category_values))
print("Sources tested:", len(sources))
```

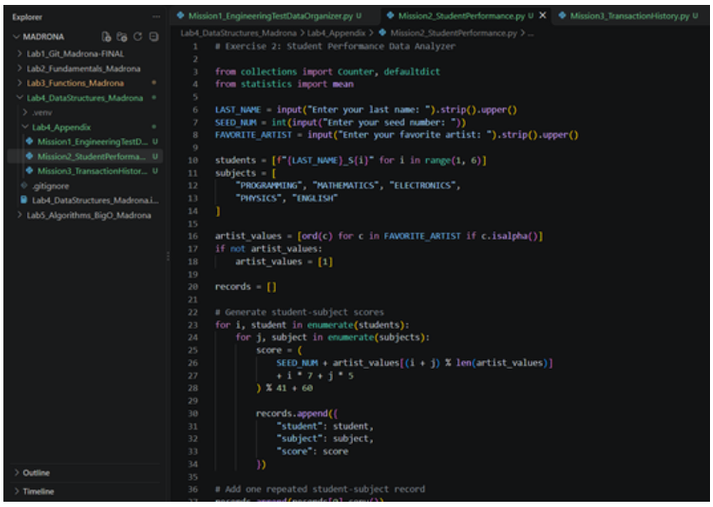








Appendix 2: Student Performance Data Analyzer



```python
# Exercise 2: Student Performance Data Analyzer

from collections import Counter, defaultdict
from statistics import mean

LAST_NAME = input("Enter your last name: ").strip().upper()
SEED_NUM = int(input("Enter your seed number: "))
FAVORITE_ARTIST = input("Enter your favorite artist: ").strip().upper()

students = [f"{LAST_NAME}_S{i}" for i in range(1, 6)]
subjects = [
    "PROGRAMMING", "MATHEMATICS", "ELECTRONICS",
    "PHYSICS", "ENGLISH"
]

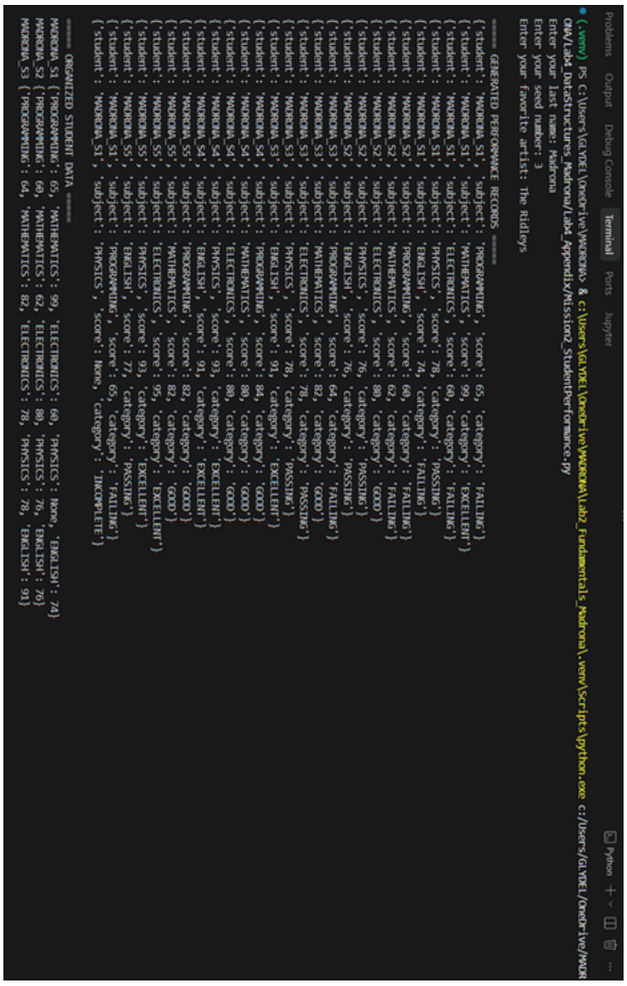
artist_values = [ord(c) for c in FAVORITE_ARTIST if c.isalpha()]
if not artist_values:
    artist_values = [1]

records = []

# Generate student-subject scores
for i, student in enumerate(students):
    for j, subject in enumerate(subjects):
        score = (
            SEED_NUM + artist_values[(i + j) % len(artist_values)]
            + i * 7 + j * 5
        ) % 41 + 60

        records.append({
            "student": student,
            "subject": subject,
            "score": score
        })

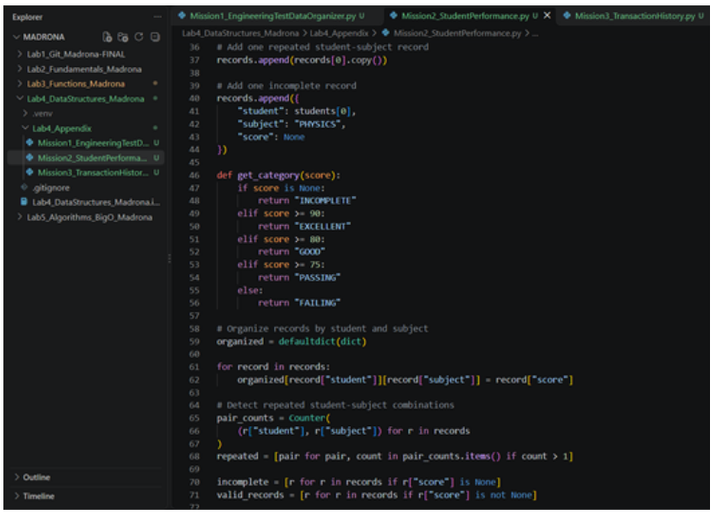
# Add one repeated student-subject record
```



```
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA\Lab2_Fundamentals_Madrona\.venv\Scripts\python.exe c:/Users/GLYDEL/OneDrive/MADRONA/Lab4_DataStructures_Madrona/Lab4_Appendix/Mission2_StudentPerformance.py
Enter your last name: Madrona
Enter your seed number: 3
Enter your favorite artist: The Ridleys

===== GENERATED PERFORMANCE RECORDS =====
{'student': 'MADRONA_S1', 'subject': 'PROGRAMMING', 'score': 65, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'MATHEMATICS', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S1', 'subject': 'ELECTRONICS', 'score': 60, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'PHYSICS', 'score': 78, 'category': 'PASSING'}
{'student': 'MADRONA_S1', 'subject': 'ENGLISH', 'score': 74, 'category': 'FAILING'}
{'student': 'MADRONA_S2', 'subject': 'PROGRAMMING', 'score': 60, 'category': 'FAILING'}
{'student': 'MADRONA_S2', 'subject': 'MATHEMATICS', 'score': 62, 'category': 'FAILING'}
{'student': 'MADRONA_S2', 'subject': 'ELECTRONICS', 'score': 80, 'category': 'GOOD'}
{'student': 'MADRONA_S2', 'subject': 'PHYSICS', 'score': 76, 'category': 'PASSING'}
{'student': 'MADRONA_S2', 'subject': 'ENGLISH', 'score': 76, 'category': 'PASSING'}
{'student': 'MADRONA_S3', 'subject': 'PROGRAMMING', 'score': 64, 'category': 'FAILING'}
{'student': 'MADRONA_S3', 'subject': 'MATHEMATICS', 'score': 82, 'category': 'GOOD'}
{'student': 'MADRONA_S3', 'subject': 'ELECTRONICS', 'score': 78, 'category': 'PASSING'}
{'student': 'MADRONA_S3', 'subject': 'PHYSICS', 'score': 78, 'category': 'PASSING'}
{'student': 'MADRONA_S3', 'subject': 'ENGLISH', 'score': 91, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S4', 'subject': 'PROGRAMMING', 'score': 84, 'category': 'GOOD'}
{'student': 'MADRONA_S4', 'subject': 'MATHEMATICS', 'score': 80, 'category': 'GOOD'}
{'student': 'MADRONA_S4', 'subject': 'ELECTRONICS', 'score': 80, 'category': 'GOOD'}
{'student': 'MADRONA_S4', 'subject': 'PHYSICS', 'score': 93, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S4', 'subject': 'ENGLISH', 'score': 91, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S5', 'subject': 'PROGRAMMING', 'score': 82, 'category': 'GOOD'}
{'student': 'MADRONA_S5', 'subject': 'MATHEMATICS', 'score': 95, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S5', 'subject': 'ELECTRONICS', 'score': 95, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S5', 'subject': 'PHYSICS', 'score': 93, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S5', 'subject': 'ENGLISH', 'score': 77, 'category': 'PASSING'}
{'student': 'MADRONA_S1', 'subject': 'PROGRAMMING', 'score': 65, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}

===== ORGANIZED STUDENT DATA =====
MADRONA_S1 {'PROGRAMMING': 65, 'MATHEMATICS': 99, 'ELECTRONICS': 60, 'PHYSICS': None, 'ENGLISH': 74}
MADRONA_S2 {'PROGRAMMING': 60, 'MATHEMATICS': 62, 'ELECTRONICS': 80, 'PHYSICS': 76, 'ENGLISH': 76}
MADRONA_S3 {'PROGRAMMING': 64, 'MATHEMATICS': 82, 'ELECTRONICS': 78, 'PHYSICS': 78, 'ENGLISH': 91}
MADRONA_S4 {'PROGRAMMING': 84, 'MATHEMATICS': 80, 'ELECTRONICS': 80, 'PHYSICS': 93, 'ENGLISH': 91}
MADRONA_S5 {'PROGRAMMING': 82, 'MATHEMATICS': 95, 'ELECTRONICS': 95, 'PHYSICS': 93, 'ENGLISH': 77}
```



```python
    # Add one repeated student-subject record
    records.append(records[0].copy())

    # Add one incomplete record
    records.append({
        "student": students[0],
        "subject": "PHYSICS",
        "score": None
    })

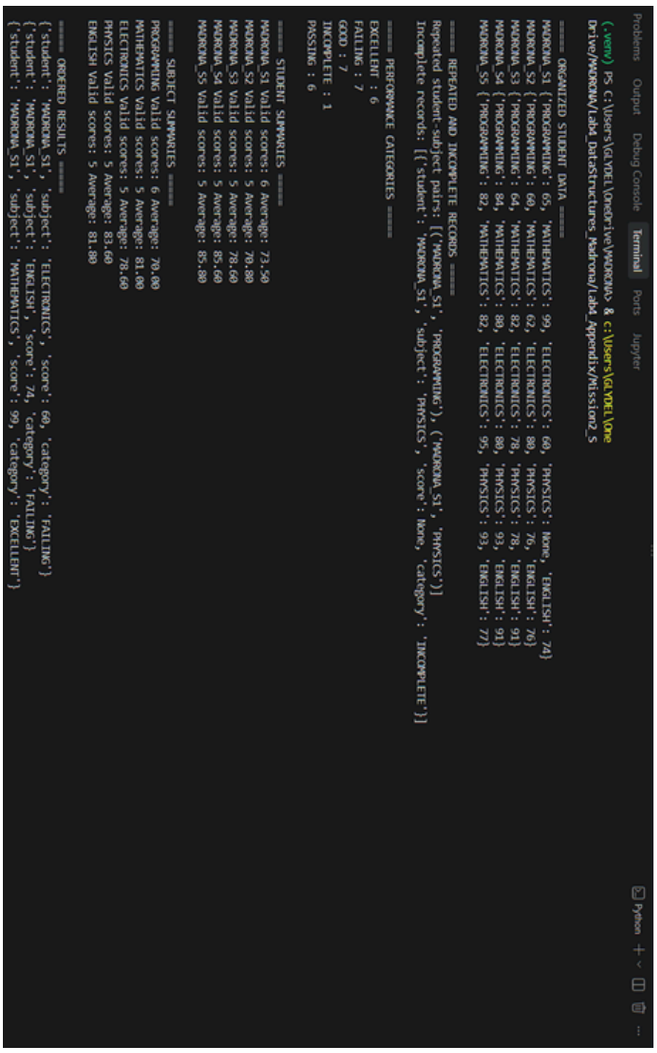
    def get_category(score):
        if score is None:
            return "INCOMPLETE"
        elif score >= 90:
            return "EXCELLENT"
        elif score >= 80:
            return "GOOD"
        elif score >= 75:
            return "PASSING"
        else:
            return "FAILING"

    # Organize records by student and subject
    organized = defaultdict(dict)

    for record in records:
        organized[record["student"]][record["subject"]] = record["score"]

    # Detect repeated student-subject combinations
    pair_counts = Counter(
        (r["student"], r["subject"]) for r in records
    )
    repeated = [pair for pair, count in pair_counts.items() if count > 1]

    incomplete = [r for r in records if r["score"] is None]
    valid_records = [r for r in records if r["score"] is not None]
```

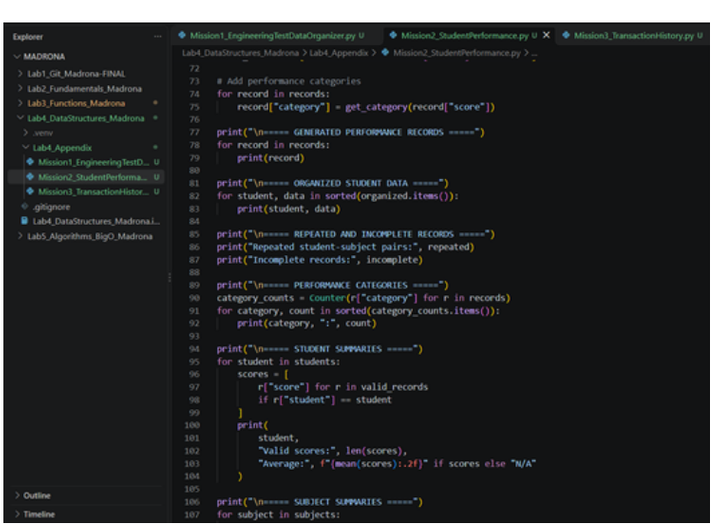


```
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA> & c:\Users\GLYDEL\OneDrive\MADRONA/Lab4_DataStructures_Madrona/Lab4_Appendix/Mission2_S

===== ORGANIZED STUDENT DATA =====
MADRONA_S1 {'PROGRAMMING': 65, 'MATHEMATICS': 99, 'ELECTRONICS': 60, 'PHYSICS': None, 'ENGLISH': 74}
MADRONA_S2 {'PROGRAMMING': 60, 'MATHEMATICS': 62, 'ELECTRONICS': 80, 'PHYSICS': 76, 'ENGLISH': 76}
MADRONA_S3 {'PROGRAMMING': 64, 'MATHEMATICS': 82, 'ELECTRONICS': 78, 'PHYSICS': 78, 'ENGLISH': 91}
MADRONA_S4 {'PROGRAMMING': 84, 'MATHEMATICS': 80, 'ELECTRONICS': 80, 'PHYSICS': 93, 'ENGLISH': 91}
MADRONA_S5 {'PROGRAMMING': 82, 'MATHEMATICS': 95, 'ELECTRONICS': 95, 'PHYSICS': 93, 'ENGLISH': 77}

===== REPEATED AND INCOMPLETE RECORDS =====
Repeated student-subject pairs: [('MADRONA_S1', 'PROGRAMMING'), ('MADRONA_S1', 'PHYSICS')]
Incomplete records: [{'student': 'MADRONA_S1', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}]

===== PERFORMANCE CATEGORIES =====
EXCELLENT : 7
FAILING : 6
GOOD : 7
INCOMPLETE : 1
PASSING : 6
```



```python

    # Add performance categories
    for record in records:
        record["category"] = get_category(record["score"])

    print("\n===== GENERATED PERFORMANCE RECORDS =====")
    for record in records:
        print(record)

    print("\n===== ORGANIZED STUDENT DATA =====")
    for student, data in sorted(organized.items()):
        print(student, data)

    print("\n===== REPEATED AND INCOMPLETE RECORDS =====")
    print("Repeated student-subject pairs:", repeated)
    print("Incomplete records:", incomplete)

    print("\n===== PERFORMANCE CATEGORIES =====")
    category_counts = Counter(r["category"] for r in records)
    for category, count in sorted(category_counts.items()):
        print(category, ":", count)

    print("\n===== STUDENT SUMMARIES =====")
    for student in students:
        scores = [
            r["score"] for r in valid_records
            if r["student"] == student
        ]
        print(
            student,
            "Valid scores:", len(scores),
            "Average:", f"{mean(scores):.2f}" if scores else "N/A"
        )

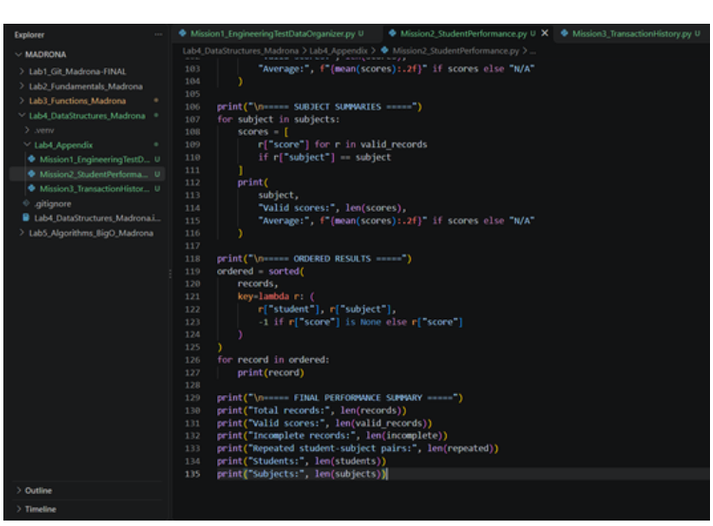
    print("\n===== SUBJECT SUMMARIES =====")
    for subject in subjects:
```



```
===== STUDENT SUMMARIES =====
MADRONA_S1 Valid scores: 6 Average: 73.50
MADRONA_S2 Valid scores: 5 Average: 70.80
MADRONA_S3 Valid scores: 5 Average: 78.60
MADRONA_S4 Valid scores: 5 Average: 85.60
MADRONA_S5 Valid scores: 5 Average: 85.80

===== SUBJECT SUMMARIES =====
PROGRAMMING Valid scores: 6 Average: 70.80
MATHEMATICS Valid scores: 5 Average: 81.80
ELECTRONICS Valid scores: 5 Average: 78.60
PHYSICS Valid scores: 5 Average: 83.60
ENGLISH Valid scores: 5 Average: 81.80

===== ORDERED RESULTS =====
{'student': 'MADRONA_S1', 'subject': 'ELECTRONICS', 'score': 60, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'ENGLISH', 'score': 74, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'MATHEMATICS', 'score': 99, 'category': 'EXCELLENT'}
```



```python
            "Average:", f"{mean(scores):.2f}" if scores else "N/A"
        )

    print("\n===== SUBJECT SUMMARIES =====")
    for subject in subjects:
        scores = [
            r["score"] for r in valid_records
            if r["subject"] == subject
        ]
        print(
            subject,
            "Valid scores:", len(scores),
            "Average:", f"{mean(scores):.2f}" if scores else "N/A"
        )

    print("\n===== ORDERED RESULTS =====")
    ordered = sorted(
        records,
        key=lambda r: (
            r["student"], r["subject"],
            -1 if r["score"] is None else r["score"]
        )
    )
    for record in ordered:
        print(record)

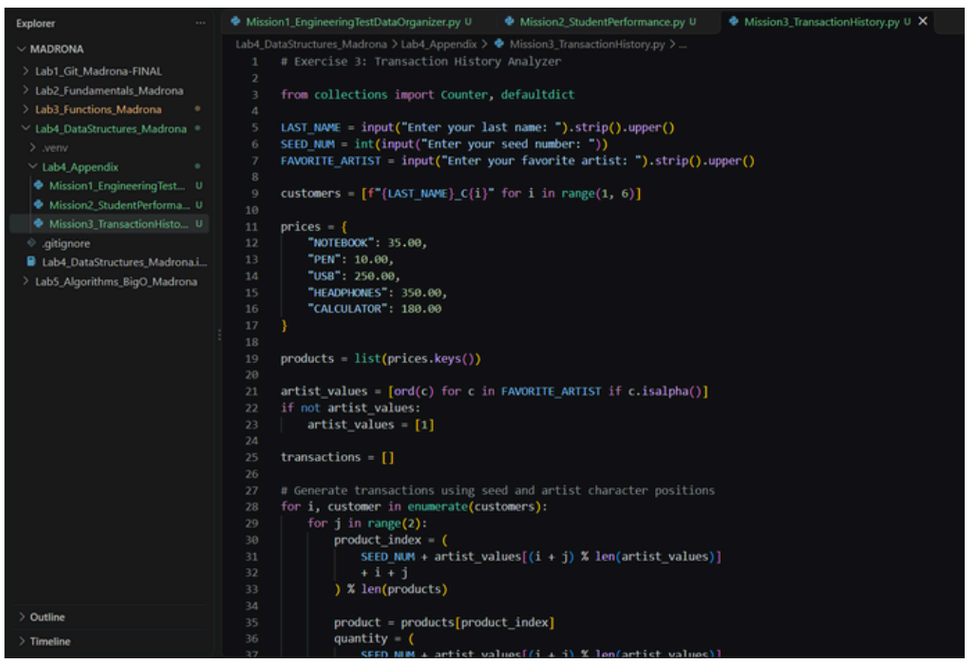
    print("\n===== FINAL PERFORMANCE SUMMARY =====")
    print("Total records:", len(records))
    print("Valid scores:", len(valid_records))
    print("Incomplete records:", len(incomplete))
    print("Repeated student-subject pairs:", len(repeated))
    print("Students:", len(students))
    print("Subjects:", len(subjects))
```

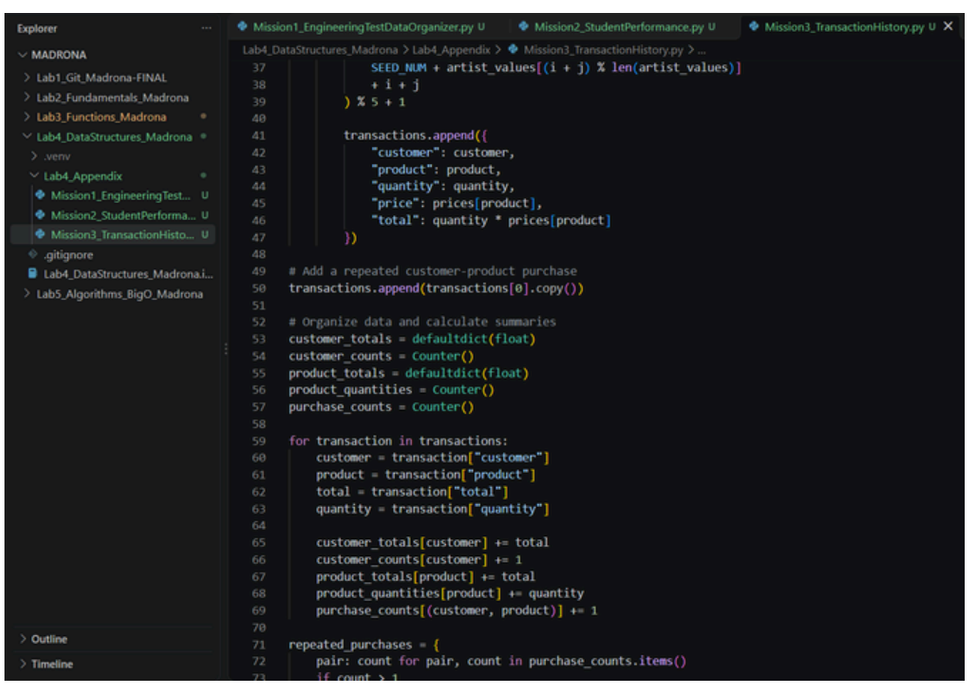
```
Problems  Output  Debug Console  Terminal  Ports  Jupyter                                                    Python

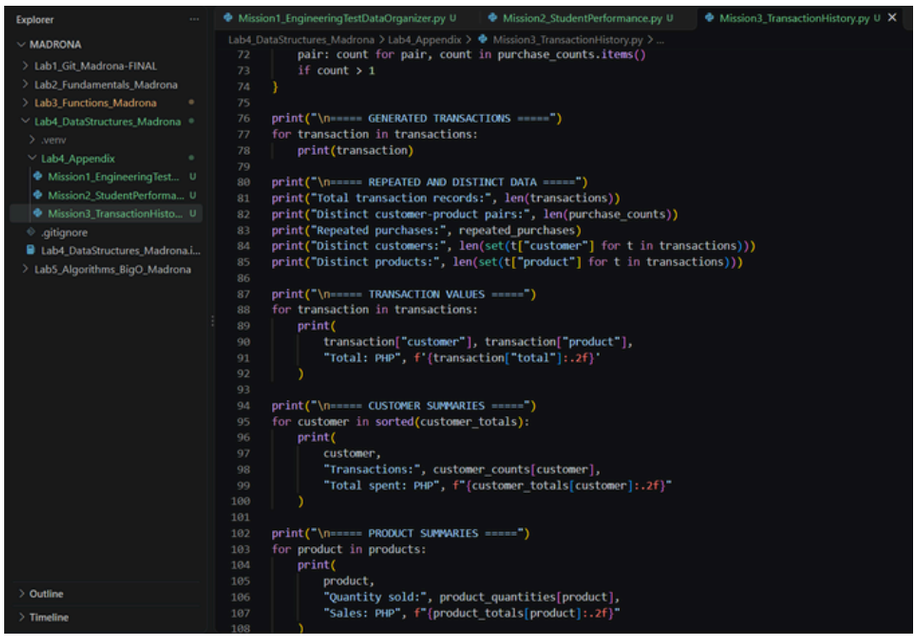
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA> & c:\Users\GLYDEL\One
Drive/MADRONA/Lab4_DataStructures_Madrona/Lab4_Appendix/Mission2_S
===== ORDERED RESULTS =====
{'student': 'MADRONA_S1', 'subject': 'ELECTRONICS', 'score': 60, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'ENGLISH', 'score': 74, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'MATHEMATICS', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S1', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}
{'student': 'MADRONA_S1', 'subject': 'PHYSICS', 'score': 78, 'category': 'PASSING'}
{'student': 'MADRONA_S1', 'subject': 'PROGRAMMING', 'score': 65, 'category': 'FAILING'}
{'student': 'MADRONA_S1', 'subject': 'PROGRAMMING', 'score': 65, 'category': 'FAILING'}
{'student': 'MADRONA_S2', 'subject': 'ELECTRONICS', 'score': 80, 'category': 'GOOD'}
{'student': 'MADRONA_S2', 'subject': 'ENGLISH', 'score': 76, 'category': 'PASSING'}
{'student': 'MADRONA_S2', 'subject': 'MATHEMATICS', 'score': 62, 'category': 'FAILING'}
{'student': 'MADRONA_S2', 'subject': 'PHYSICS', 'score': 76, 'category': 'PASSING'}
{'student': 'MADRONA_S2', 'subject': 'PROGRAMMING', 'score': 60, 'category': 'FAILING'}
{'student': 'MADRONA_S3', 'subject': 'ELECTRONICS', 'score': 78, 'category': 'PASSING'}
{'student': 'MADRONA_S3', 'subject': 'ENGLISH', 'score': 91, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S3', 'subject': 'MATHEMATICS', 'score': 82, 'category': 'GOOD'}
{'student': 'MADRONA_S3', 'subject': 'PHYSICS', 'score': 78, 'category': 'PASSING'}
{'student': 'MADRONA_S3', 'subject': 'PROGRAMMING', 'score': 64, 'category': 'FAILING'}
{'student': 'MADRONA_S4', 'subject': 'ELECTRONICS', 'score': 80, 'category': 'GOOD'}
{'student': 'MADRONA_S4', 'subject': 'ENGLISH', 'score': 91, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S4', 'subject': 'MATHEMATICS', 'score': 80, 'category': 'GOOD'}
{'student': 'MADRONA_S4', 'subject': 'PHYSICS', 'score': 93, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S4', 'subject': 'PROGRAMMING', 'score': 84, 'category': 'GOOD'}
{'student': 'MADRONA_S5', 'subject': 'ELECTRONICS', 'score': 95, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S5', 'subject': 'ENGLISH', 'score': 77, 'category': 'PASSING'}
{'student': 'MADRONA_S5', 'subject': 'MATHEMATICS', 'score': 82, 'category': 'GOOD'}
{'student': 'MADRONA_S5', 'subject': 'PHYSICS', 'score': 93, 'category': 'EXCELLENT'}
{'student': 'MADRONA_S5', 'subject': 'PROGRAMMING', 'score': 82, 'category': 'GOOD'}

===== FINAL PERFORMANCE SUMMARY =====
Total records: 27
Valid scores: 26
Incomplete records: 1
Repeated student-subject pairs: 2
Students: 5
Subjects: 5
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA>
```

Appendix 3: Transaction History Analyzer







```
    pair: count for pair, count in purchase_counts.items()
    if count > 1
}

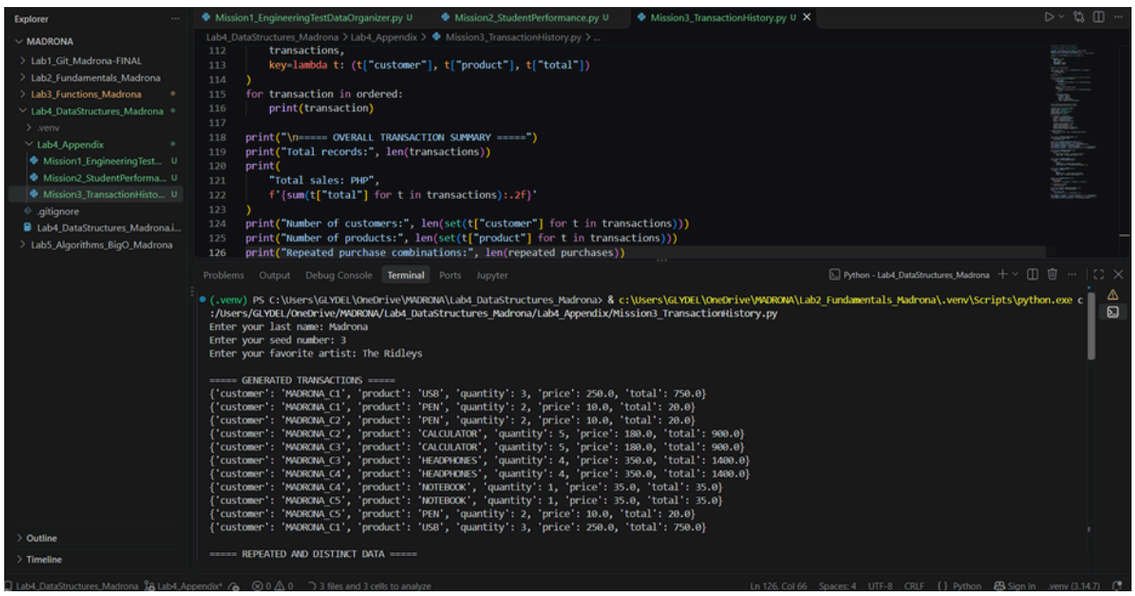
print("\n===== GENERATED TRANSACTIONS =====")
for transaction in transactions:
    print(transaction)

print("\n===== REPEATED AND DISTINCT DATA =====")
print("Total transaction records:", len(transactions))
print("Distinct customer-product pairs:", len(purchase_counts))
print("Repeated purchases:", repeated_purchases)
print("Distinct customers:", len(set(t["customer"] for t in transactions)))
print("Distinct products:", len(set(t["product"] for t in transactions)))

print("\n===== TRANSACTION VALUES =====")
for transaction in transactions:
    print(
        transaction["customer"], transaction["product"],
        "Total: PHP", f"{transaction["total"]:.2f}"
    )

print("\n===== CUSTOMER SUMMARIES =====")
for customer in sorted(customer_totals):
    print(
        customer,
        "Transactions:", customer_counts[customer],
        "Total spent: PHP", f"{customer_totals[customer]:.2f}"
    )

print("\n===== PRODUCT SUMMARIES =====")
for product in products:
    print(
        product,
        "Quantity sold:", product_quantities[product],
        "Sales: PHP", f"{product_totals[product]:.2f}"
    )
```



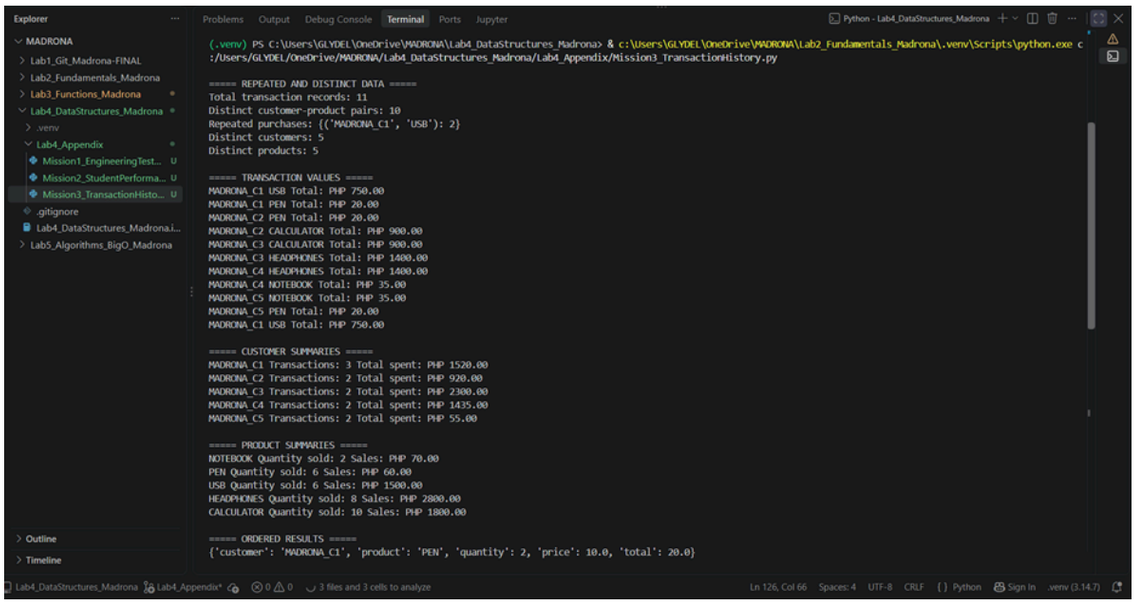
```
    transactions,
    key=lambda t: (t["customer"], t["product"], t["total"])
)
for transaction in ordered:
    print(transaction)

print("\n===== OVERALL TRANSACTION SUMMARY =====")
print("Total records:", len(transactions))
print(
    "Total sales: PHP",
    f'{sum(t["total"] for t in transactions):.2f}'
)
print("Number of customers:", len(set(t["customer"] for t in transactions)))
print("Number of products:", len(set(t["product"] for t in transactions)))
print("Repeated purchase combinations:", len(repeated_purchases))
```

```
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA\Lab4_DataStructures_Madrona> & c:\Users\GLYDEL\OneDrive\MADRONA\Lab2_Fundamentals_Madrona\.venv\Scripts\python.exe c
:/Users/GLYDEL/OneDrive/MADRONA/Lab4_DataStructures_Madrona/Lab4_Appendix/Mission3_TransactionHistory.py
Enter your last name: Madrona
Enter your seed number: 3
Enter your favorite artist: The Ridleys

===== GENERATED TRANSACTIONS =====
{'customer': 'MADRONA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'MADRONA_C1', 'product': 'PEN', 'quantity': 2, 'price': 10.0, 'total': 20.0}
{'customer': 'MADRONA_C2', 'product': 'PEN', 'quantity': 2, 'price': 10.0, 'total': 20.0}
{'customer': 'MADRONA_C2', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'MADRONA_C3', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'MADRONA_C3', 'product': 'HEADPHONES', 'quantity': 4, 'price': 350.0, 'total': 1400.0}
{'customer': 'MADRONA_C4', 'product': 'HEADPHONES', 'quantity': 4, 'price': 350.0, 'total': 1400.0}
{'customer': 'MADRONA_C4', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'MADRONA_C5', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'MADRONA_C5', 'product': 'PEN', 'quantity': 2, 'price': 10.0, 'total': 20.0}
{'customer': 'MADRONA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}

===== REPEATED AND DISTINCT DATA =====
```



```
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA\Lab4_DataStructures_Madrona> & c:\Users\GLYDEL\OneDrive\MADRONA\Lab2_Fundamentals_Madrona\.venv\Scripts\python.exe c
:/Users/GLYDEL/OneDrive/MADRONA/Lab4_DataStructures_Madrona/Lab4_Appendix/Mission3_TransactionHistory.py

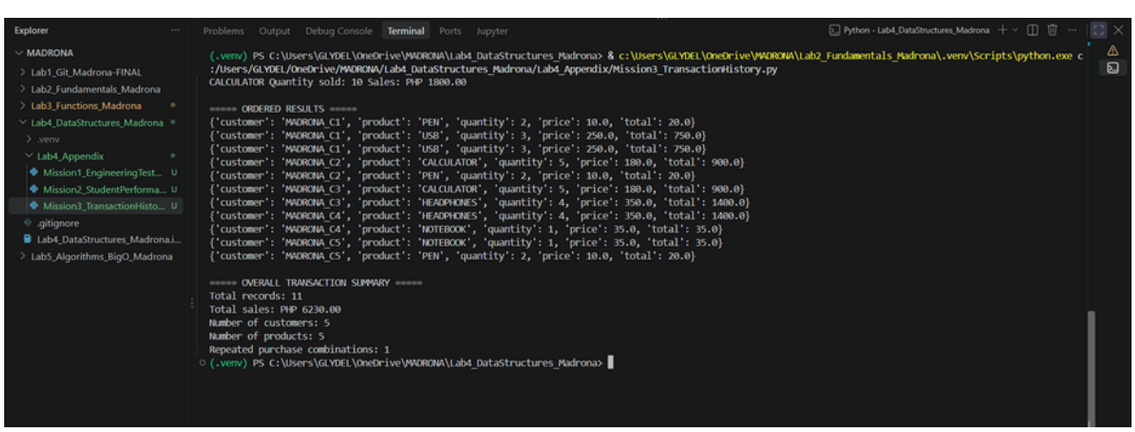
===== REPEATED AND DISTINCT DATA =====
Total transaction records: 11
Distinct customer-product pairs: 10
Repeated purchases: {('MADRONA_C1', 'USB'): 2}
Distinct customers: 5
Distinct products: 5

===== TRANSACTION VALUES =====
MADRONA_C1 USB Total: PHP 750.00
MADRONA_C1 PEN Total: PHP 20.00
MADRONA_C2 PEN Total: PHP 20.00
MADRONA_C2 CALCULATOR Total: PHP 900.00
MADRONA_C3 CALCULATOR Total: PHP 900.00
MADRONA_C3 HEADPHONES Total: PHP 1400.00
MADRONA_C4 HEADPHONES Total: PHP 1400.00
MADRONA_C4 NOTEBOOK Total: PHP 35.00
MADRONA_C5 NOTEBOOK Total: PHP 35.00
MADRONA_C5 PEN Total: PHP 20.00
MADRONA_C1 USB Total: PHP 750.00

===== CUSTOMER SUMMARIES =====
MADRONA_C1 Transactions: 3 Total spent: PHP 1520.00
MADRONA_C2 Transactions: 2 Total spent: PHP 920.00
MADRONA_C3 Transactions: 2 Total spent: PHP 2300.00
MADRONA_C4 Transactions: 2 Total spent: PHP 1435.00
MADRONA_C5 Transactions: 2 Total spent: PHP 55.00

===== PRODUCT SUMMARIES =====
NOTEBOOK Quantity sold: 2 Sales: PHP 70.00
PEN Quantity sold: 6 Sales: PHP 60.00
USB Quantity sold: 6 Sales: PHP 1500.00
HEADPHONES Quantity sold: 8 Sales: PHP 2800.00
CALCULATOR Quantity sold: 10 Sales: PHP 1800.00

===== ORDERED RESULTS =====
{'customer': 'MADRONA_C1', 'product': 'PEN', 'quantity': 2, 'price': 10.0, 'total': 20.0}
```



```
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA\Lab4_DataStructures_Madrona> & c:\Users\GLYDEL\OneDrive\MADRONA\Lab2_Fundamentals_Madrona\.venv\Scripts\python.exe c
:/Users/GLYDEL/OneDrive/MADRONA/Lab4_DataStructures_Madrona/Lab4_Appendix/Mission3_TransactionHistory.py
CALCULATOR Quantity sold: 10 Sales: PHP 1800.00

===== ORDERED RESULTS =====
{'customer': 'MADRONA_C1', 'product': 'PEN', 'quantity': 2, 'price': 10.0, 'total': 20.0}
{'customer': 'MADRONA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'MADRONA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'MADRONA_C2', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'MADRONA_C2', 'product': 'PEN', 'quantity': 2, 'price': 10.0, 'total': 20.0}
{'customer': 'MADRONA_C3', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'MADRONA_C3', 'product': 'HEADPHONES', 'quantity': 4, 'price': 350.0, 'total': 1400.0}
{'customer': 'MADRONA_C4', 'product': 'HEADPHONES', 'quantity': 4, 'price': 350.0, 'total': 1400.0}
{'customer': 'MADRONA_C4', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'MADRONA_C5', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'MADRONA_C5', 'product': 'PEN', 'quantity': 2, 'price': 10.0, 'total': 20.0}

===== OVERALL TRANSACTION SUMMARY =====
Total records: 11
Total sales: PHP 6230.00
Number of customers: 5
Number of products: 5
Repeated purchase combinations: 1
(.venv) PS C:\Users\GLYDEL\OneDrive\MADRONA\Lab4_DataStructures_Madrona>
```